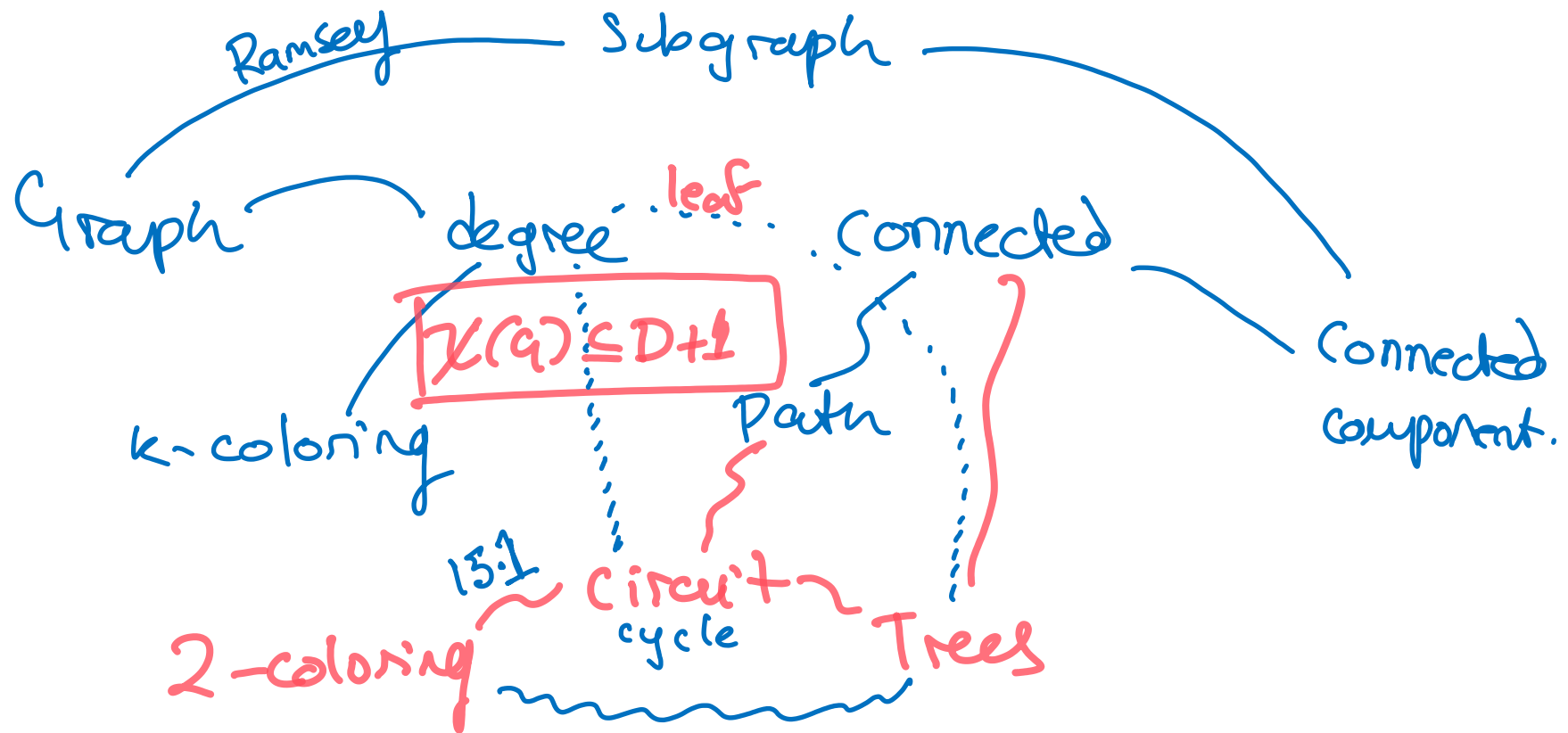


Lecture 16

So far:



I Properties of Trees

II Spanning Trees

III Cycles

Theorem 16.1: If $T = (V, E)$ is a tree on n vertices then $|E| = n - 1$.

Proof:

By induction on the number of vertices.

Let $P(n)$ = Thm^{16.1} Statement for trees with n vertices,

Base Case: If $n=1$, T is an isolated vertex with $0=1-1$ edges, so $P(1)$ holds.

Inductive Step: Assume $k \geq 1$. Assume $P(k)$.

Let $T = (V, E)$ be a tree with $k+1$ vertices.

Let $x \in V$ be a leaf of T and let

$T' = T - x$. Let y be the unique vertex

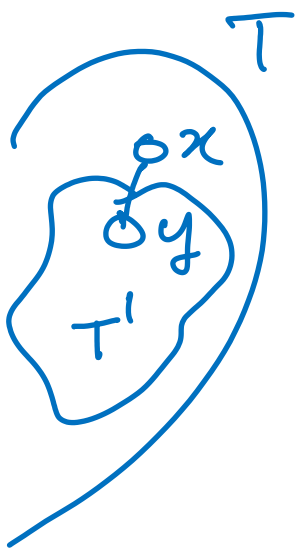
adjacent to x in T . By Lec 15, T' is

a tree. Since $T' = (V', F)$ has k vertices, $P(k)$

implies that $|F| = k-1$. Since $E = F \cup \{xy\}$

and $xy \notin F$, we conclude $|E| = (k-1) + 1 = k$,

as desired. $\square$



Corollary: ①

"maximal
acyclic"

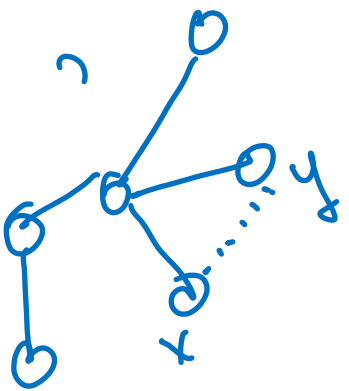
If $T = (V, E)$ is a tree and $x, y \in V$ with $xy \notin E$ then $\underbrace{T + xy}_{\text{add edge}}$ contains a cycle.

②

"minimal
connected"

If $T = (V, E)$ is a tree and $xy \in E$ then $\underbrace{T - xy}_{\text{delete an edge}}$ is disconnected.

Proof:



① Let T be a tree on n vertices.
Let $xy \notin E$ and $T' = T + xy$. Since T and T' have the same vertex set, T' has n vertices. Observe that T' is connected since T is connected and adding an edge cannot remove any paths. Since T' has n edges, it cannot be a tree by Thm 6.1-. Thus, T' has a cycle.

(2)

<HW>.

→ Claim: T' is connected.

Proof: Let a, b be two distinct vertices in T' . Since T is connected, we can choose a path γ from a to b in T . Since $T \subseteq T'$, γ is a path in T' , as desired. $\square$

Math 54 :

acyclic $\longleftrightarrow$ linearly indep.

connected $\longleftrightarrow$ span

both : basis $\longrightarrow$ has # elements
= $\dim(V)$.

Last time:

G is 2 colorable $\overset{15.1}{\longleftrightarrow}$ G has no odd circuit
 $\underset{(HW)}{\longleftrightarrow}$ G has no odd cycle subgraph

Theorem 16.2: Every tree is 2-colorable.

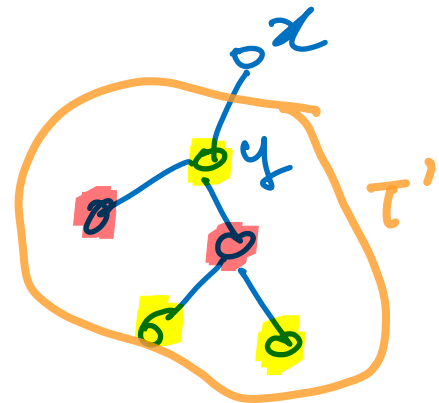
Proof #1: If T is a tree, it does not contain an odd cycle as a subgraph. Since it is acyclic. So by Thm 15.1, it is 2-colorable.

Proof #2: By induction on the number of vertices.
Let $P(n)$ = Thm 16.2 for trees with n vertices.

Base Case: If $n=1$, T is 1-colorable.

Inductive Step: Assume $k \geq 1$ and $P(k)$.

Let $T = (V, E)$ be a tree with $k+1$ vertices. Choose a leaf $x \in V$ with unique neighbor y in T . Let $T' = T - x = (W, F)$. By $P(k)$,



we can choose a 2-coloring
 $f: W \rightarrow \{1, 2\}$ of T' .

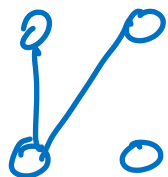
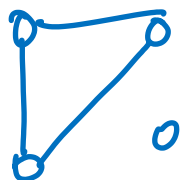
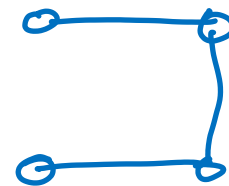
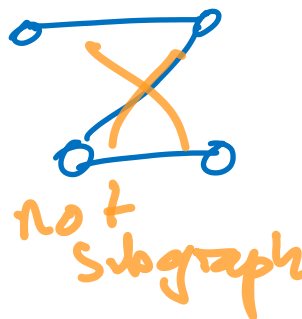
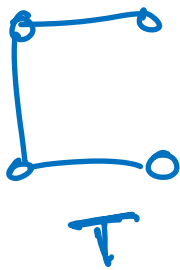
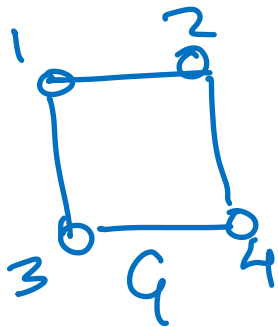
Extend f to a function $g: V \rightarrow \{1, 2\}$
by
$$\begin{cases} g(a) = f(a) & \text{if } a \neq x \\ g(x) \text{ is different from } g(y). \end{cases}$$

g is a 2-coloring of T' and of the
edge xy , so g is a 2-coloring of T . $\square$

II Spanning Trees

Defn: If $G = (V, E)$ is a graph, a spanning tree
of G is a tree $T = (V, F)$ which is
a subgraph of G .

ex:



not connected.

not connected

no spanning tree.

Theorem 16.3: A graph is connected iff it contains a spanning tree.

Proof:

($\leftarrow$) (HW).

($\rightarrow$) By induction on the number of edges.

Let $P(m)$ = "Every connected graph with m edges contains a spanning tree."

We will show this for all $m \geq 0$.

Base Case: If $m=0$, G is an isolated vertex
and $T=G$ is a spanning tree of G .

Inductive Step: Assume $m \geq 0$. Assume $P(m)$.

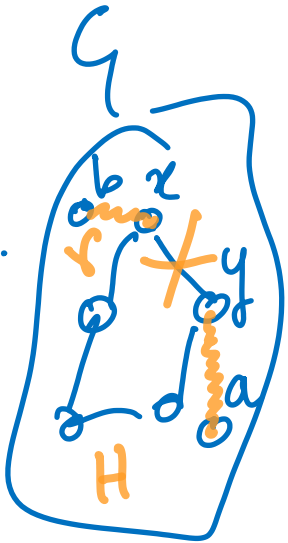
Let $G=(V,E)$ be ^{connected} arbitrary with $|E|=m+1$.

If G is acyclic, G is a tree so $T=G$ suffices.

If G is not a tree, we can choose a
cycle C in G . Choose an edge

$xy \in C$ and let $H = G - xy = (V,F)$.

We will show H is connected. Let $a, b \in V$
be distinct vertices in H .



Choose a path γ from a to b in G .

Let γ' be the path obtained by replacing every occurrence of xy in γ by the path traversing C from x to y which avoids the edge xy .

γ' is a path from a to b in H

Since it does not traverse xy , so H is connected. H has M edges, so by $P(m)$ we can choose a spanning tree T of H . Since H has the same vertices as G , T is a spanning tree of G , as desired. $\square$

Remark: Repeatedly finding cycles and deleting edges will produce a tree.

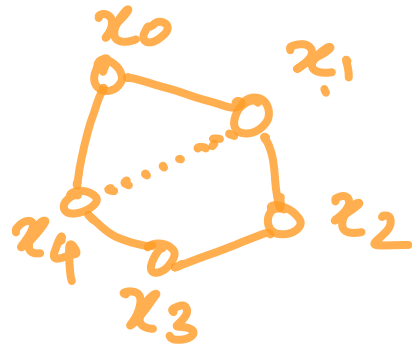
All inductive graph theory proofs give algorithms for constructing the desired coloring, tree, etc.

III ~~Cycles~~ . Isomorphism

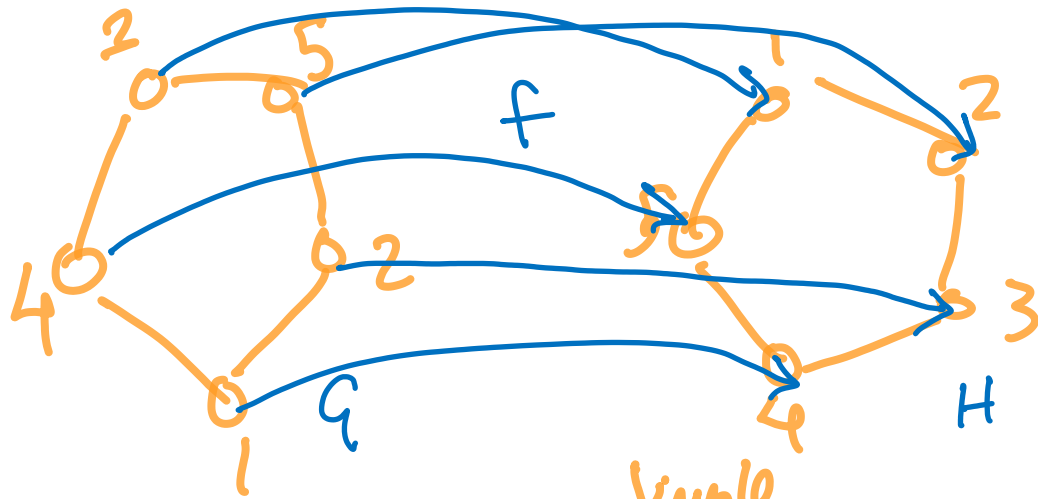
Prob:

The # of ^{simple, connected} graphs on $\{1, \dots, n\}$ ^{up to relabelling.} with $\deg(v) = 2$ for all $v \in \{1, \dots, n\}$ is —

$n = 5$:



• in such a graph, every vertex is contained in a cycle.

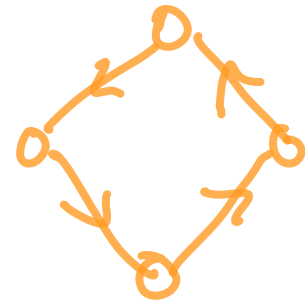
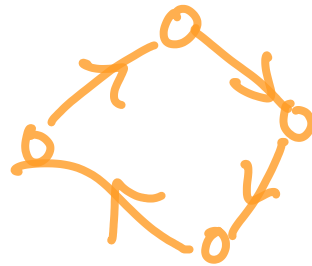


• if connected, there is exactly one cycle.

(B) 1: ^{simple} connected, up to relabeling.

(D) ≥ 2 : drop any of these.

(C) 2:



cycle.

2 orientations for a circuit.

Defn: Two graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic if there is a bijection $f: V \rightarrow W$

Such that for all $x, y \in V$

$$xy \in E \iff f(x)f(y) \in F.$$